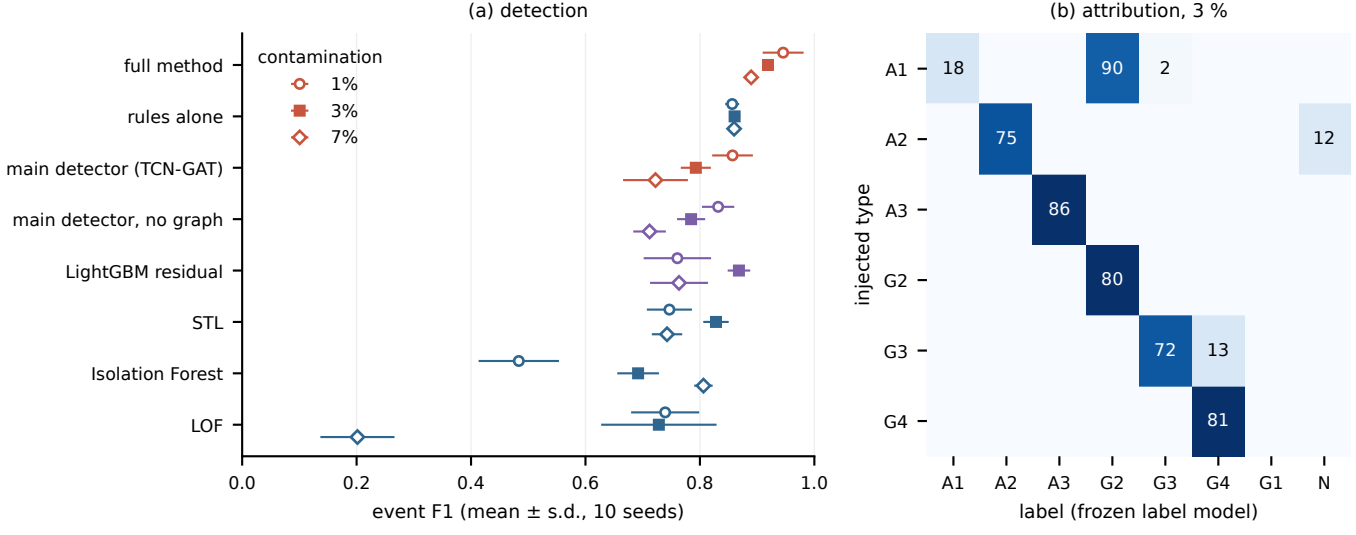
(a) detection
full method
rules alone
main detector (TCN-GAT)
main detector, no graph
LightGBM residual
STL
Isolation Forest
LOF
contamination
1%
3%
7%
0.0
0.2
0.4
0.6
0.8
1.0
event F1 (mean ± s.d., 10 seeds)
(b) attribution, 3 %
injected type
A1
A2
A3
G2
G3
G4
18
90
2
75
12
86
80
72
13
81
A1
A2
A3
G2
G3
G4
G1
N
label (frozen label model)